

HW/Lab 1: Modulation, Upconversion, BER

ECSE 4560/6560: Modern Communication Systems

RPI Fall 2025

Goals and Instructions

The goals for this assignment are:

- Convert passband waveforms to complex baseband and back.
- Derive BER for BPSK and QPSK under AWGN.
- Understand spectra of Amplitude modulated signals and the effect of carrier-frequency offset.
- Practice FFT-based spectral estimation, frequency-axis labeling, and power scaling.

Follow these instructions:

- **Collaboration:** You can discuss concepts, but write your own solutions and code.
- **Tools:** We will primarily use MATLAB (and/or Python). Directions use standard MATLAB functions. *Strongly recommended* toolboxes: Signal Processing, Communications. (Base-MATLAB fallbacks are acceptable.)
- **Submission:** upload a single PDF with your solutions and MATLAB code to Gradescope before the deadline. You may use MATLAB's **Publish** feature to generate a compiled PDF and attach it as your single submission. Include your name on the PDF.

Theory Problems

1. Complex Baseband Representation

The transmitted signal from a radio has the following form

$$\tilde{s}(t) = \text{Re}\{s(t)e^{j\omega_c t}\}$$

where $\tilde{s}(t)$ is the passband signal, $s(t)$ is corresponding baseband signal, and $\omega_c = 2\pi f_c$ is the radian frequency for up-conversion.

Consider the following passband BPSK signal:

$$\tilde{s}(t) = A \cos(\omega_c t + a_i \pi)$$

and QPSK signals

$$\tilde{s}(t) = A \cos(\omega_c t + a_i \pi) - A \sin(\omega_c t + b_i \pi)$$

where A is a constant amplitude and a_i and b_i are independent binary random variables that can take on values $a_i, b_i \in \{0, 1\}$.

- a) Find the complex baseband representation $s(t)$ of the *BPSK signal*.
- b) Find the complex baseband representation $s(t)$ of the *QPSK signal*.

2. BER, theoretical expression for BPSK, QPSK

Now, take the $s(t)$ defined in the above question. In a practical setting, the received signal, $r(t)$ will include both the transmitted signal and noise.

$$r(t) = s(t) + n(t)$$

Where $n(t)$ can be viewed as a realization of a Gaussian random variable with zero mean and variance σ^2 . Note for complex $n(t)$, we have

$$n(t) = n_I(t) + jn_Q(t), \quad (1)$$

$$n_I(t), n_Q(t) \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad \sigma^2 = \frac{N_0}{2}. \quad (2)$$

Assume that $s(t)$ sends bits 0, 1 each in an equi-probable 1/2 probability. Also assume that bit is decoded from $r(t)$ by observing the phase $\angle r(t)$ and declaring bit 0 is $-\frac{\pi}{2} \leq \angle r(t) \leq \frac{\pi}{2}$, and bit 1 otherwise.

- a) Compute conditional probabilities $p_{1|0}$ and $p_{0|1}$. Here $p_{1|0}$ denotes the probability that bit 0 was sent but instead bit 1 is decoded by observing $\angle r(t)$ instead. $p_{0|1}$ likewise denotes the probability that bit 1 was sent but bit 0 is decoded from $\angle r(t)$.
- b) Use this to compute probability of error: $p_e = \frac{1}{2}p_{1|0} + \frac{1}{2}p_{0|1}$.
- c) (ECSE 6560 Only) Provide a sketch solution to compute p_e for QPSK. Why are p_e for QPSK and BPSK the same?

3. Spectrum of AM modulated signals

Let $s(t)$ be a bandlimited baseband signal with a frequency content $S(f)$ given by $1 - |f|/f_{\max}$, where f is the frequency and $f_{\max}$ is the maximum frequency of the baseband signal. This baseband signal is multiplied by $\cos(2\pi f_c t)$ to produce an amplitude-modulated passband signal $\tilde{s}(t) = s(t) \cos(2\pi f_c t)$, where f_c is the carrier frequency and $f_c \gg f_{\max}$.

- a) Sketch the amplitude of the frequency spectrum of the baseband signal $S(f)$.
- b) Sketch the amplitude of the frequency spectrum of the amplitude-modulated passband signal $\tilde{S}(f)$.
- c) The AM signal is down-converted by multiplying $\tilde{s}(t)$ by a coherent signal of the form $\cos(2\pi f_c t)$. The signal is filtered by an ideal lowpass filter with a cutoff frequency $f_{\max}$. Sketch the amplitude of the frequency spectrum of the down-converted signal.

d) Frequency Down-conversion errors

Here we consider the effect of a frequency error in the down-conversion. An AM signal $\tilde{s}(t) = s(t) \cos(2\pi f_c t)$ is down-converted using $\cos(2\pi f_c [1 + x]t)$ where x is a relative frequency error. Sketch the amplitude of the frequency spectrum of the down-converted signal for: (a) $x = 0$, (b) $x = f_{\max}/(10f_c)$, and (c) $x = f_{\max}/f_c$. Comment on the results. Does a frequency down-conversion error cause distortion at Nyquist rate of the baseband signal? Explain.

Matlab Simulations

1. Spectral Analysis using the FFT

Generate a sinusoid time-series vector $x(n)$ at frequency $f_c = 256\text{Hz}$, sampled at rate $f_s = 1/T_s = 8192\text{Hz}$.

$$x(n) = x(nT_s) = A \cos(2\pi f_c n T_s)$$

- a) Using the generated time-series $x(n)$, apply the 32-point FFT to create the vector $X[k]$.

$$X[k] = \text{FFT}\{x(n)\}$$

- b) Generate the power sequence of $X[k]$ denoted $P[k] = (X[k])^2$. You can do this by typing `abs(X[k])^2`.
- c) Plot $P[k]$ vs frequency in hertz. (Note that the frequency axis from left to right corresponds to $f \in [0, 2\pi]$ in normalized frequency units)
- d) Plot the properly frequency shifted version of $P[k]$ using the Matlab function `fftshift(·)`. (This will use a frequency axis with limits $f \in [-\pi, \pi]$ in normalized frequency units).
- e) Using your results from part (d), correctly label the frequency axis using non-normalized frequency units in hertz. (i.e., the two delta impulses corresponding to the Fourier transform of a sinusoid should lie on frequencies -256Hz and $+256\text{Hz}$). *Hint: What is the frequency spacing in hertz between each point of the FFT output?*
- f) Using your results from part (e), plot the power spectrum in decibels scale using the `db(·)` function.

$$\text{db}(x) = 10 \log_{10}(x)$$

Deliverable: Include the plots generated from parts (c), (d), and (f). Note: Figures must include titles, labeled axes with units, and legends when relevant.

2. Power Spectrum

Following the steps from Simulation Problem 1, generate and plot the power spectra (on a decibel scale) of the following three waveforms using at least 1024 points:

- a) *Random Sequence of -1's and +1's (BPSK - Binary Phase Shift Keying)*

First, generate a random vector of -1's and +1's:

- By using the function `randsrc(1, 128)`. This will create a random vector of -1's and +1's of size 128.
- Convert the random sequences of -1's and +1's into rectangular pulses of 8 samples wide, all with the same value by using the command `kron(x, ones(1, 8))`. You should now have a vector of size 1024.

- b) *Random Sequence of 0's and 1's (ASK - Amplitude Shift Keying)*

Modify your time series from part (a) to be a random sequence of 0's and 1's.

- c) *Additive "White" Gaussian Noise (AWGN)*

Gaussian random vector with unit variance. (Use the function `randn(1, 1024)`).

Deliverable: Include the three power spectra from parts (a)-(c).

Note: Figures must include titles, labeled axes with units, and legends when relevant.